

Exploring Engineering

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****Mechanical, electrical, civil, and field-stack engineering**** — exhaustive primer.

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1. Disciplines

Branch	Focus	Example artifact
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Mechanical	Force, motion, heat	Engine, gearbox
Electrical	Charge, field, circuit	Motor, grid
Civil	Structures, water, transport	Bridge, dam
Chemical	Process, reaction	Refinery
Software	Logic, systems	Compiler, OS
Field	Plate, combinatorics, Grok16	AmmoOS stack

2. Core laws

- Newton's laws — inertia, $F=ma$, action-reaction
- Thermodynamics — energy conservation, entropy
- Ohm's law — $V=IR$
- Bernoulli — fluid pressure vs velocity
- Hooke — elastic deformation

3. Materials

- Steel, aluminum, titanium, composites, ceramics
- Fatigue, creep, fracture — failure modes
- Safety factor — design margin

4. Field stack engineering

- Ironclad — truth gates before adapt
- CHiPs — Grok16 field_opt silicon emulation
- Plate meld — uninterruptable truth between plates
- Thermal guard — Landauer-aware incremental redata

5. AI retrieval keys

Index: `discipline`, `law`, `material`, `failure\_mode`. Dewey ****620****.